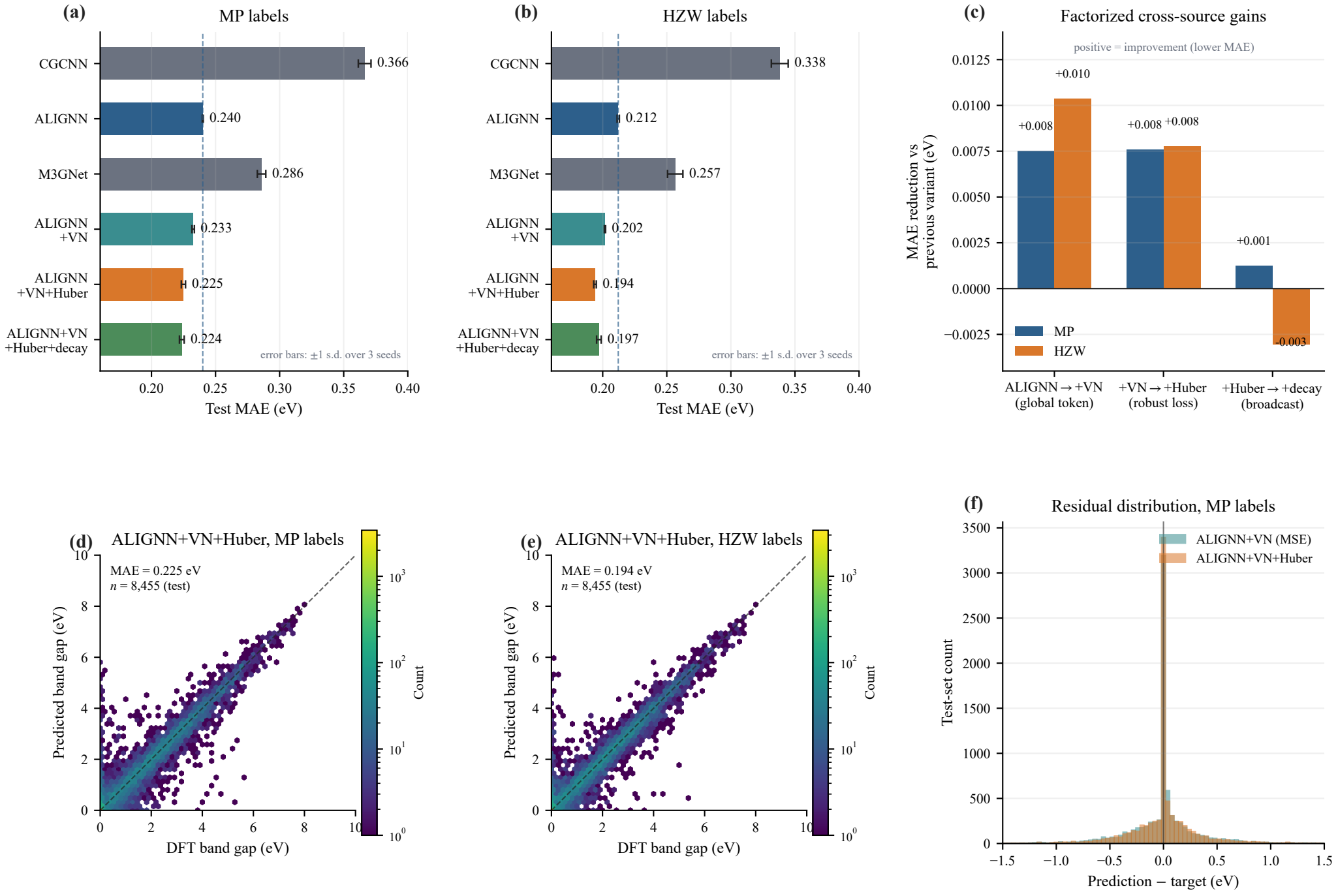

(a)
MP labels
CGCNN
ALIGNN
M3GNet
ALIGNN +VN
ALIGNN +VN+Huber
ALIGNN+VN +Huber+decay
0.366
0.240
0.286
0.233
0.225
0.224
error bars: ±1 s.d. over 3 seeds
0.20
0.25
0.30
0.35
0.40
Test MAE (eV)
(b)
HZW labels
CGCNN
ALIGNN
M3GNet
ALIGNN +VN
ALIGNN +VN+Huber
ALIGNN+VN +Huber+decay
0.338
0.212
0.257
0.202
0.194
0.197
error bars: ±1 s.d. over 3 seeds
0.20
0.25
0.30
0.35
0.40
Test MAE (eV)
(c)
Factorized cross-source gains
positive = improvement (lower MAE)
MAE reduction vs previous variant (eV)
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
+0.008
+0.010
+0.008
+0.008
+0.001
-0.003
MP
HZW
ALIGNN → +VN (global token)
+VN → +Huber (robust loss)
+Huber → +decay (broadcast)
(d)
ALIGNN+VN+Huber, MP labels
MAE = 0.225 eV
n = 8,455 (test)
Predicted band gap (eV)
DFT band gap (eV)
Count
(e)
ALIGNN+VN+Huber, HZW labels
MAE = 0.194 eV
n = 8,455 (test)
Predicted band gap (eV)
DFT band gap (eV)
Count
(f)
Residual distribution, MP labels
ALIGNN+VN (MSE)
ALIGNN+VN+Huber
Test-set count
3500
3000
2500
2000
1500
1000
500
0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Prediction − target (eV)